

Linear Partial Differential equation of order one.

An equation involving partial differential coefficients of a function of two or more variable is known as a partial differential equation. We use the following notations for partial differential equation.

$$p = \frac{\partial z}{\partial x}, q = \frac{\partial z}{\partial y}, r = \frac{\partial^2 z}{\partial x^2}, s = \frac{\partial^2 z}{\partial x \partial y}, t = \frac{\partial^2 z}{\partial y^2}$$

* Derivation of a PDE by the elimination of arbitrary constants.

* u

functions.

* Eliminate arbitrary constants & function from the following equation to form the PDE.

$$1. z = \left(ax + \frac{y}{a} + b\right)^2$$

$$2. z = Ae^{pt} \cdot \sin px.$$

$$3. z = a(x+y) + b(x-y) + abt.$$

$$4. z = axe^y + \frac{1}{2} a^2 e^y + b.$$

$$5. z = y^2 + 2f\left(\frac{1}{x} + \ln y\right)$$

$$6. z = f(x+iy) + F(x-iy), i^2 = -1$$

$$7. z = f(x-at) + F(x+at)$$

$$8. z = f(xy) + g\left(\frac{x}{y}\right).$$

Soln: 1. Given,

$$Z = \left(ax + \frac{y}{a} + b\right)^2$$

Differentiate, Partially w.r. to x & y .

$$\frac{\partial Z}{\partial x} = 2\left(ax + \frac{y}{a} + b\right) \frac{\partial}{\partial x} \left(ax + \frac{y}{a} + b\right)$$

$$\frac{\partial Z}{\partial x} = 2\left(ax + \frac{y}{a} + b\right) \cdot a \quad \therefore$$

$$\frac{\partial Z}{\partial y} = 2\left(ax + \frac{y}{a} + b\right) \frac{\partial}{\partial y} \left(ax + \frac{y}{a} + b\right)$$

$$\frac{\partial Z}{\partial y} = 2\left(ax + \frac{y}{a} + b\right) \cdot \frac{1}{a}$$

Again,

$$\frac{\partial^2 Z}{\partial x^2} = 2a \cdot a$$

$$= 2a^2 \quad \text{①}$$

&

$$\frac{\partial^2 Z}{\partial y^2} = \frac{2}{a} \cdot \frac{1}{a}$$
$$= \frac{2}{a^2}$$

$$\Rightarrow \frac{\partial^2 z}{\partial y^2} = 2 \left(\frac{2}{\partial^2 z}{\partial x^2} \right) \text{ [Using (i)]}$$

$$\Rightarrow \frac{\partial^2 z}{\partial y^2} = \frac{4}{\partial^2 z}{\partial x^2}$$

$$\therefore \left(\frac{\partial^2 z}{\partial y^2} \right) \left(\frac{\partial^2 z}{\partial x^2} \right) = 4 \quad \underline{\text{Ans}}$$

Soln:-2 Given,

$$z = A e^{pt} \sin px$$

$$\frac{\partial z}{\partial x} = A p e^{pt} \cos px$$

$$\frac{\partial^2 z}{\partial x^2} = -A p^2 e^{pt} \sin px \quad \text{--- (i)}$$

&

$$\frac{\partial z}{\partial t} = A p e^{pt} \sin px$$

$$\frac{\partial^2 z}{\partial t^2} = A p^2 e^{pt} \sin px \quad \text{--- (ii)}$$

(i) + (ii)

$$\frac{\partial^2 z}{\partial x^2} + \frac{\partial^2 z}{\partial t^2} = -Ap^2 e^{pt} \sin px + Ap^2 e^{pt} \sin px$$

$$\frac{\partial^2 z}{\partial x^2} + \frac{\partial^2 z}{\partial t^2} = 0$$

Ans

Soln:-3 Given,

$$z = a(x+y) + b(x-y) + abt + c$$

Differentiate Partially w.r. to x & y .

$$\frac{\partial z}{\partial x} = a + b$$

$$\frac{\partial z}{\partial y} = a - b$$

$$\frac{\partial z}{\partial t} = ab$$

We know,

$$ab = \left(\frac{a+b}{2} \right)^2 - \left(\frac{a-b}{2} \right)^2$$

$$\therefore 4 \frac{\partial z}{\partial t} = \left(\frac{\partial z}{\partial x} \right)^2 - \left(\frac{\partial z}{\partial y} \right)^2$$

Ans

Soln:-4 Given,

$$z = ax e^y + \frac{1}{2} a^2 e^y + b$$

$$\frac{\partial z}{\partial x} = a e^y$$

$$\Rightarrow a = e^y \frac{\partial z}{\partial x} \quad \text{--- (i)}$$

$$\frac{\partial z}{\partial y} = a x e^y + \frac{1}{2} a^2 e^y$$

$$\Rightarrow \frac{\partial z}{\partial y} = \left(e^y \frac{\partial z}{\partial x} \right) x e^y + \frac{1}{2} \left(e^y \frac{\partial z}{\partial x} \right)^2 e^y$$

$$\therefore \frac{\partial z}{\partial y} = x \frac{\partial z}{\partial x} + \frac{1}{2} \left(\frac{\partial z}{\partial x} \right)^2 e^y$$

Ans

we know

$$\left(\frac{d+0}{16} \right) - \left(\frac{d+0}{16} \right) = d$$

$$\left(\frac{16}{16} \right) - \left(\frac{16}{16} \right) = 16 \quad \therefore$$

Soln :- 5 Given,

$$z = y^2 + 2f\left(\frac{1}{x} + \ln y\right)$$

Differentiate Partially w.r. to x & y .

$$\frac{\partial z}{\partial x} = 2f'\left(\frac{1}{x} + \ln y\right) \frac{\partial}{\partial x} \left(\frac{1}{x} + \ln y\right)$$

$$\frac{\partial z}{\partial x} = 2f'\left(\frac{1}{x} + \ln y\right) \cdot \left(-\frac{1}{x^2}\right) \quad \text{--- (i)}$$

$$\begin{aligned} \& \frac{\partial z}{\partial y} &= 2y + 2f'\left(\frac{1}{x} + \ln y\right) \cdot \frac{\partial}{\partial y} \left(\frac{1}{x} + \ln y\right) \\ &= 2y + 2f'\left(\frac{1}{x} + \ln y\right) \cdot \left(\frac{1}{y}\right) \end{aligned}$$

$$\Rightarrow \frac{\partial z}{\partial y} = 2y + \left(-x^2 \frac{\partial z}{\partial x}\right) \cdot \left(\frac{1}{y}\right) \quad [\text{using (i)}]$$

$$\Rightarrow \frac{\partial z}{\partial y} = 2y - \frac{x^2}{y} \left(\frac{\partial z}{\partial x}\right)$$

$$\therefore q = 2y - \frac{px^2}{y}$$

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Soln:- 6 Given, $z = f(x+iy) + F(x-iy)$

Differentiate Partially w.r. to x & y .

$$\frac{\partial z}{\partial x} = f'(x+iy) \cdot 1 + F'(x-iy) \cdot 1$$

$$= f'(x+iy) + F'(x-iy)$$

$$\frac{\partial z}{\partial y} = f'(x+iy) \cdot i + F'(x-iy) \cdot (-i)$$

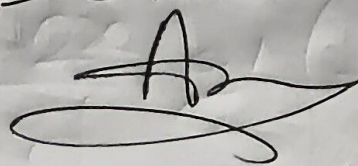
Again,

$$\frac{\partial^2 z}{\partial x^2} = f''(x+iy) + F''(x-iy) \quad \text{--- (i)}$$

$$\frac{\partial^2 z}{\partial y^2} = f''(x+iy) \cdot i^2 + F''(x-iy) \cdot (-i)^2$$
$$= i^2 [f''(x+iy) + F''(x-iy)]$$

$$\frac{\partial^2 z}{\partial y^2} = - \frac{\partial^2 z}{\partial x^2} \quad [\text{using (i)}]$$

$$\therefore \frac{\partial^2 z}{\partial x^2} + \frac{\partial^2 z}{\partial y^2} = 0$$



Soln:-7 Given, $z = f(x-at) + F(x+at)$

Differentiate Partially w.r. to x & t .

$$\frac{\partial z}{\partial x} = f'(x-at) + F'(x+at)$$

$$\frac{\partial z}{\partial t} = f'(x-at)(-a) + F'(x+at) \cdot a$$

Again,

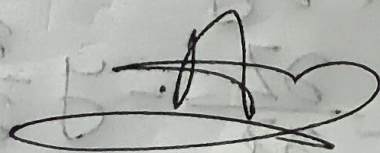
$$\frac{\partial^2 z}{\partial x^2} = f''(x-at) + F''(x+at) \quad \text{--- (i)}$$

$$\frac{\partial^2 z}{\partial t^2} = f''(x-at)(a^2) + F''(x+at) \cdot a^2$$

$$\Rightarrow \frac{\partial^2 z}{\partial t^2} = a^2 [f''(x-at) + F''(x+at)]$$

$$\Rightarrow \frac{\partial^2 z}{\partial t^2} = a^2 \frac{\partial^2 z}{\partial x^2}$$

$$\therefore a^2 \frac{\partial^2 z}{\partial x^2} = \frac{\partial^2 z}{\partial t^2}$$



Soln:-8 Given,

$$z = f(xy) + g\left(\frac{x}{y}\right)$$

$$\frac{\partial z}{\partial x} = f'(xy) \cdot y + g'\left(x/y\right) \cdot \frac{1}{y} \quad \text{--- (i)}$$

$$\frac{\partial z}{\partial y} = f'(xy) \cdot x + g'\left(x/y\right) \cdot \left(-\frac{x}{y^2}\right) \quad \text{--- (ii)}$$

$$\frac{\partial^2 z}{\partial x^2} = y^2 f''(xy) + \frac{1}{y^2} g''\left(x/y\right) \quad \text{--- (iii)}$$

$$\frac{\partial^2 z}{\partial y^2} = x^2 f''(xy) + \frac{x^2}{y^4} g''\left(x/y\right) + \frac{2x}{y^3} g'\left(x/y\right) \quad \text{--- (iv)}$$

$$\underline{x \times (i) - y \times (ii)}$$

$$x \frac{\partial z}{\partial x} = xy f'(xy) + \frac{x}{y} g'\left(\frac{x}{y}\right)$$

$$y \left(\frac{\partial z}{\partial y} \right) = xy f'(xy) - \frac{x}{y} g'\left(\frac{x}{y}\right)$$

$$\therefore x \frac{\partial z}{\partial x} - y \frac{\partial z}{\partial y} = \frac{2x}{y} g'\left(\frac{x}{y}\right) \quad \text{--- (v)}$$

$$y^2 \times \textcircled{iv} - x^2 \times \textcircled{iii}$$

$$x^2 \frac{\partial^2 z}{\partial x^2} = x^2 y^2 \cdot f''(xy) + \frac{x^2}{y^2} \cdot g''\left(\frac{x}{y}\right)$$

$$y^2 \frac{\partial^2 z}{\partial y^2} = x^2 y^2 \cdot f''(xy) + \frac{x^2}{y^2} \cdot g''\left(\frac{x}{y}\right) + \frac{2x}{y} g'\left(\frac{x}{y}\right).$$

$$\therefore y^2 \frac{\partial^2 z}{\partial y^2} - x^2 \frac{\partial^2 z}{\partial x^2} = \frac{2x}{y} \cdot g'\left(\frac{x}{y}\right) - \textcircled{vi}$$

from (v) & (vi),

$$y^2 \frac{\partial^2 z}{\partial y^2} - x^2 \frac{\partial^2 z}{\partial x^2} = x \frac{\partial z}{\partial x} - y \frac{\partial z}{\partial y}$$

$$\Rightarrow x^2 \frac{\partial^2 z}{\partial x^2} - y^2 \frac{\partial^2 z}{\partial y^2} + x \frac{\partial z}{\partial x} - y \frac{\partial z}{\partial y} = 0$$

$$\therefore x^2 r - y^2 t + x p - y q = 0.$$

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